

A.4 HIP with Output-Layer-Only Adaptation

We also test whether HIP can be approximated by changing only the final mapping from hidden states to logits. Starting from the Llama3-8B and Qwen3-8B base checkpoints, we keep the same paired paraphrase dataset $\mathcal{D}$ and the same 10-round evaluation protocol, but freeze the entire model except for the output layer `lm_head`. This ablation therefore trains a single output projection rather than the standard LoRA-based paraphraser used in the main paper. We then evaluate the resulting models on the same 256-example evaluation set with the same semantic, GPTZero, and Pangram metrics.

As Fig. 9 shows, output-layer-only adaptation does not reproduce the main HIP trade-off. For Llama3-8B, the detector curves move in the right direction, but only at the cost of a much sharper drop in semantic preservation than under standard HIP. For Qwen3-8B, the failure is stronger: semantic scores remain relatively high compared to that in Llama3-8B, but detector-assigned human probability stays far below the HIP curve on both GPTZero and Pangram metrics.

Taken together, these results suggest that the main HIP effect cannot be explained as a simple output-layer transformation, such as reweighting or suppressing a small set of suspicious tokens. At least for the models studied here, successful humanization appears to require changes deeper in the representation pathway than the final logit layer alone.

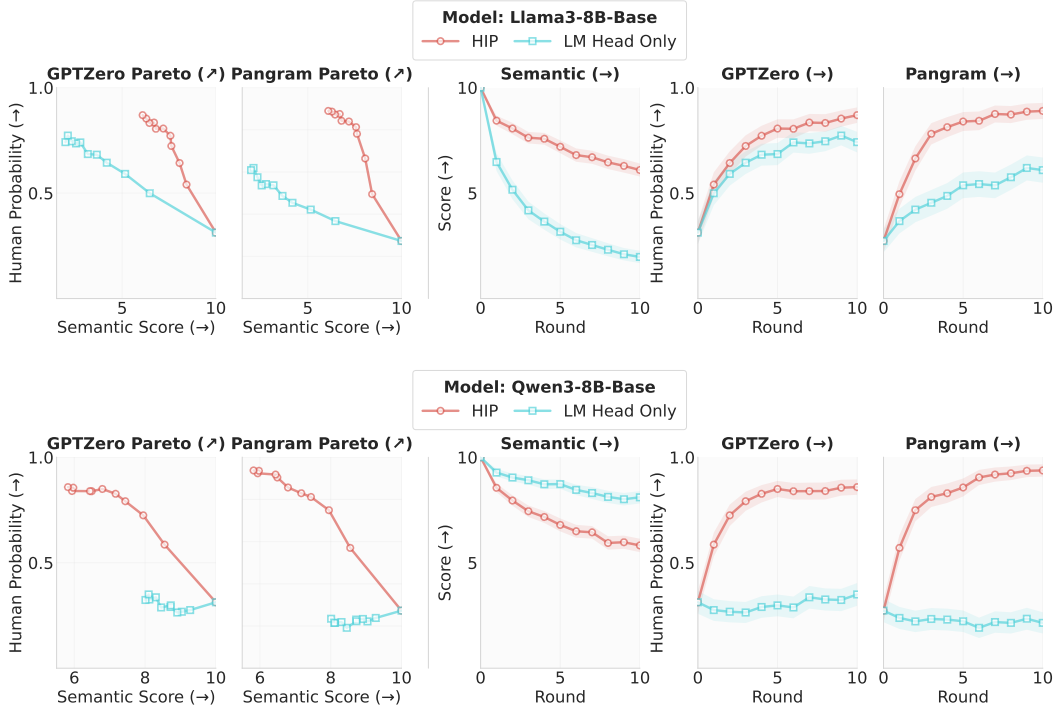


Figure 9: Standard HIP versus output-layer-only adaptation. Top: Llama3-8B-Base. Bottom: Qwen3-8B-Base. In each panel, the first two subplots show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic preservation and human-likeness, while the last three show semantic score, GPTZero human probability, and Pangram human probability over the rounds, respectively. Shaded areas indicate 95% confidence intervals. Output-layer-only adaptation underperforms standard HIP in both families, and especially on Qwen3-8B it fails to produce a useful semantic-evasion trade-off.

B Qualitative Examples

To help interpret the quantitative metrics in the main paper, this appendix presents matched qualitative examples for HIP on Llama3-8B and for all four baseline methods: Simple Paraphrase, DIPPER, SilverSpeak, and StealthRL. The five examples appear in the same source-text order for every method and are sampled in an unbiased manner. For HIP, Simple Paraphrase, DIPPER, and StealthRL, each example shows the original AI text together with the first, second, and tenth rewrites, along with their semantic, GPTZero, and Pangram scores. SilverSpeak is a single-pass method, so its examples use a one-row layout with the original text and the transformed output. For readability in the PDF, the displayed SilverSpeak outputs are ASCII-normalized renderings of the homoglyph-substituted text, while the reported scores are computed on the original attacked text.

B.1 HIP on Llama3-8B

Example 1.

Original AI Text	MAGE / XSum	HIP (Llama3-8B) Round 1	MAGE / XSum
Semantic: 10 (ref.) GPTZero: 99.9% Pangram: 0.0%		Semantic: 10 GPTZero: 99.9% Pangram: 0.0%	
Amputee football is played all over the world but it is in its infancy in Scotland. Amputee football is a form of the game that is adapted for people who have had one or both legs amputated. The Scottish Amputee Football Association (SAFA) is the governing body for the sport in Scotland and is working to set up a new league. There are currently only two teams in Scotland - Glasgow Rangers and Heart of Midlothian - but SAFA is hopeful that more clubs will join in the future. Amputee football is a fast-paced and physically demanding sport, but it is also great fun and can be enjoyed by people of all ages and abilities. If you are interested in finding out more about amputee football in Scotland, or if you would like to join a team, please contact SAFA.		Amputee football is a sport that is played all over the world, but it is still in its infancy in Scotland. Amputee football is a variation of the sport that has been adapted for people who have had their legs amputated. The Scottish Amputee Football Association (SAFA) is the governing body for amputee football in Scotland and is in the process of setting up a new league. There are currently only two teams in Scotland (Glasgow Rangers and Heart of Midlothian) but SAFA hopes that more teams will join in the future. Amputee football is a fast paced, physically demanding sport, but it is also a lot of fun and can be enjoyed by people of all ages and abilities. If you are interested in finding out more about amputee football in Scotland or if you are interested in joining a team, please contact SAFA.	
HIP (Llama3-8B) Round 2	MAGE / XSum	HIP (Llama3-8B) Round 10	MAGE / XSum
Semantic: 10 GPTZero: 99.9% Pangram: 100.0%		Semantic: 9 GPTZero: 100.0% Pangram: 100.0%	
Amputee football is a sport that is played all over the world, but it is in its infancy in Scotland. Amputee football is a variation of the game that has been adapted for people who have had their legs amputated. The Scottish Amputee Football Association (SAFA) is the governing body for amputee football in Scotland and are currently setting up a new league. There are currently only 2 teams in Scotland, Glasgow Rangers and Heart of Midlothian, but SAFA are hoping to get more teams in the future. Amputee football is a fast and physically demanding sport, but it is also a lot of fun and can be played by people of all ages and abilities. If you are interested in finding out more about amputee football in Scotland or if you are interested in joining a team please get in touch with SAFA.		Amputee Football is played all over the world, including Scotland. Amputee football is just football, but it's amputees playing the game. At the moment, the Scottish Amputee Football Association (SAFA) are running a league for this game. There are only 2 teams in Scotland at the moment, Glasgow Rangers and Heart of Midlothian. The Scottish Amputee Football Association are trying to run more teams for this game as soon as possible. Amputee football is a fast, physical game. But it's a fun game and can be played by anyone of any age or ability. If you would like any more information on Amputee Football in Scotland, or if you would like to sign up to one of our teams, then please contact the Scottish Amputee Football Association.	

Example 2.

Original AI Text	RAID / Abstracts	HIP (Llama3-8B) Round 1	RAID / Abstracts
Semantic: 10 (ref.) GPTZero: 0.0% Pangram: 0.0%		Semantic: 10 GPTZero: 0.0% Pangram: 0.0%	
In this paper, we investigate the eigenvalues of non-Hermitian random matrices and the Brown measure of non-normal operators using a Hermitian reduction and linearization method. We show that the eigenvalues of non-Hermitian random matrices can be reduced to those of their Hermitian counterparts, and that the Brown measure of non-normal operators can be expressed in terms of the eigenvalue distribution of a associated linear operator. Our approach relies on a novel application of the perturbation theory of eigenvalues and the moment method, and provides a new perspective on the relationship between non-Hermitian and Hermitian systems. We demonstrate the power of our method by computing the eigenvalue distribution of several classes of non-Hermitian random matrices, including the Gaussian Unitary Ensemble (GUE), the Gaussian Orthogonal Ensemble (GOE) and the Wishart matrix ensemble. Furthermore, we apply our results to study the Brown measure of non-normal operators, which has applications in various fields such as signal processing, control theory, and random matrix theory. Our work provides a significant generalization of previous results in the field, and sheds light on the universal properties of non-Hermitian systems. The techniques developed here are expected to have far-reaching implications for the study of non-Hermitian phenomena in physics, engineering and other areas where complex systems are prevalent.	In this paper, we study the eigenvalues of non-Hermitian random matrices and the Brown measure of non-normal operators by using a technique of Hermitian reduction and linearization. We show that the eigenvalues of non-Hermitian random matrices can be reduced to those of their Hermitian counterparts, and the Brown measure of non-normal operators can be expressed by the eigenvalue distribution of a linear operator. Our method is based on a new application of the perturbation theory of eigenvalues and the moment method, and provides a new angle to the relation between non-Hermitian and Hermitian systems. We apply our method to calculate the eigenvalue distribution of some non-Hermitian random matrices, including the Gaussian Unitary Ensemble (GUE), the Gaussian Orthogonal Ensemble (GOE), and the Wishart matrix ensemble. Furthermore, we apply our results to study the Brown measure of non-normal operators, which has applications in various areas such as signal processing, control theory, and random matrix theory. Our results provide a significant generalization of the previous works and provide a deeper understanding of the universal properties of non-Hermitian systems. Our approach has significant implications for understanding non-Hermitian phenomena in physics, engineering, and other areas where complex systems occur.		
HIP (Llama3-8B) Round 2	RAID / Abstracts	HIP (Llama3-8B) Round 10	RAID / Abstracts
Semantic: 9 GPTZero: 0.0% Pangram: 0.0%		Semantic: 8 GPTZero: 99.9% Pangram: 0.0%	
In this paper, we study the eigenvalues of non-Hermitian random matrices and the Brown measure of non-normal operators by a method of Hermitian reduction and linearization. We show that the eigenvalues of non-Hermitian random matrices can be reduced to the eigenvalues of their Hermitian counterparts and the Brown measure of non-normal operators can be expressed in terms of the eigenvalue distribution of a linear operator. Our method is based on a new application of the perturbation theory of eigenvalues and the moment method, and shows a new direction to the relationship between non-Hermitian and Hermitian systems. Our method is applied to evaluate the eigenvalue distribution of some non-Hermitian random matrices such as Gaussian Unitary Ensemble (GUE), Gaussian Orthogonal Ensemble (GOE) and Wishart matrix ensemble. In addition, our results are applied to evaluate the Brown measure of non-normal operators, which has many applications in fields including signal processing, control theory and random matrix theory. Our results give a substantial extension of the known results, and provide a new insight into the universal properties of non-Hermitian systems. Our results have significant implications on the studies of non-Hermitian phenomenon in physics, engineering and other fields where complex systems exist.	The eigenvalues of non-Hermitian random matrices and the Brown measure of a non-normal operator are studied via the reduction to Hermitian matrices and linearization methods. The results are applicable to the study of the eigenvalues of non-Hermitian random matrices via reduction to Hermitian matrices and the study of the Brown measure of a non-normal operator via linearization of the eigenvalues of a linear operator. In addition, this method is applicable to the study of the perturbation of the eigenvalues and moment theory of non-Hermitian and Hermitian systems, which is our main application. In particular, this method is used in studying the eigenvalues of non-Hermitian random matrices, such as Gaussian Unitary Ensemble (GUE), Gaussian Orthogonal Ensemble (GOE) and Wishart matrix. Besides, the distribution of the Brown measure of a non-normal operator is studied and it has important applications in signal processing, control theory and random matrix theory. This result is much improved and can provide new ideas and insights to the study of universal properties of non-Hermitian systems. These results are used in the non-Hermitian in physics, engineering, complex systems and other fields.		